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Electron effective mass and nonparabolicity in InGaAs/InAlAs quantum wells lattice-matched to InP

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Abstract

Nonparabolic electron effective masses in InGaAs quantum wells (QWs), sandwiched by thick InAlAs barriers of 0.52-eV band offset, were studied in normal and parallel directions to the QW plane. The Normal mass was experimentally obtained by observing interband photocurrent spectra of undoped InGaAs multi-QW structures. The mass increased by more than 50% from the bulk band edge mass, $0.041m_0$. Electron eigenenergies were calculated in QWs based on Kane's three-level band theory. The calculated 'apparent' normal mass as a function of kinetic energy up to 0.5 eV agreed well with experiments. The parallel mass in n-type modulation-doped InGaAs QWs was experimentally obtained by pulse cyclotron resonance up to 100 T. The analysis in quantizing magnetic fields, modified for two-dimensional QWs, fits well with cyclotron energy. The 'apparent' parallel mass as a function of energy was obtained consistently. © 2001 Elsevier Science B.V. All rights reserved.

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1. Introduction

Understanding electronic states in quantum wells (QWs) is important for nano-structure physics and its application to wavelength design in lasers. The $In_xGa_{1-x}As(x=0.53)/In_yAl_{1-y}As(y=0.52)$ multi-quantum well (MQWs) structures lattice-matched to InP crystal are most important for application to 1.5-μm band optical telecommunication devices.

So far, the nonparabolicity of conduction band was studied mostly in narrow-gap semiconductors like InSb. The strong confinement of wavefunctions in QWs, however, facilitates the study of the non-parabolic nature of bulk conduction band, because of

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the stronger two-dimensional (2D) exciton absorption and multiplication of subband eigenstates.

In the $In_xGa_{1-x}As(x=0.53)/InP$ MQW system with a small conduction-band offset of 0.22 eV, it was reported that the observed anomalous heavy electron mass was due to the penetration of wavefunctions into InP barrier [1].

In this paper, we treat InGaAs QWs in the $In_xGa_{1-x}As(x=0.53)/In_yAl_{1-y}As(y=0.52)$ system, having a large conduction-band offset of 0.52 eV. A series of eigenenergies in photocurrent spectra of undoped InGaAs QWs was measured and used for comparison with nonparabolic band theory. 'Apparent' normal mass up to $0.07\,m_0$ was observed in a wide energy range of 0.5 eV. The 2D band calculation in a narrow-gap semiconductor agreed quantitatively with experimental results. Apparent cyclotron mass in n-type-modulation-doped InGaAs QW was also measured up to 100 T, to obtain 'apparent' parallel mass. Apparent parallel mass up to $0.06\,m_0$ was observed below 0.2 eV. Calculation and fitting gave the nonparabolic energy dependence of parallel mass.

2. Experimental procedure

For photocurrent spectroscopy, specimens called V706 and V748 were grown on n-type doped InP (1 0 0) surface by MBE. Above n-InAlAs buffer layer, undoped MQW layer including 25 or 16 MQWs was formed for V706 or V748. Above p-InAlAs layer, a p-InGaAs contact layer was laid on top of the structures for photocurrent measurement. In MQWs, thicknesses of InGaAs well and InAlAs buffer were 10 and 10 nm for V706 and 20 and 10 nm for V748. X-ray measurement ensured MQW periods of 18.8 and 31.1 nm, respectively.

The estimated In-content, X, was 0.537 and 0.536, respectively. Photocurrent was measured by scanning wavelength from HR320 monochromator with a Halogen lamp, mostly at room temperature.

For pulse cyclotron resonance, specimens called M10L were grown on InP (1 0 0) surface. Above InAlAs buffer layer, an n-type modulation-doped MQW layer including 40 MQWs was formed. Dopant was Si and nominal carrier concentration per one QW was $5\times10^{11}\,cm^{-2}$. In MQWs, thicknesses of InGaAs well and InAlAs buffer were 10 and 14 nm, respectively. X-ray measurement revealed MQW period of 23.8 nm which agreed with nominal thickness. Estimated In-content, X, was 0.535.

Cyclotron resonance was measured by scanning magnetic field at fixed wavelengths of 9.25, 9.55. 10.23, 10.59, 16.9, 28, and 119 μm, mostly at 20–30 K. Magnetic field was parallel with the light path. Magnetic field pulse, whose duration was 0.6 μs, was shot during the time when laser light pulse was on. Electron-sensitive circularly polarized light was used near 10.6 μm range.

3. Definition of masses

Apparent normal mass in a square QW was calculated in the following manner. Eigenenergies in a square-well potential is given by a formula derived from boundary conditions at the interface, combined with a set of one-dimensional Scrödinger equations in a well and a barrier as given by Eqs. (3.93) and (3.96) of Ref. [2]. Normally, effective masses in a well, $m^*_\perp$, and in a barrier, $m^*_{b\perp}$, are assumed constant. One can obtain a consistent $m^*_\perp$ from a series of observed eigenenergies in QWs.

In our experiment, however, we gave a constant $m^*_{b\perp}$ and assumed an energy-dependent $m^*_\perp(E)$ where E is the kinetic energy. Then, we got a nonparabolic electron effective mass in an energy range up to 0.5 eV [3–5]. ΔE_c was also found to be 0.52 eV. In our calculation, we got $m^*_\perp(E)$ with the same procedure after a series of eigenenergies was obtained. We call such mass as the 'apparent' normal effective mass, in experiment and theory.

Apparent parallel mass in very high magnetic fields was defined as follows. Cyclotron resonance in magnetic field is an optical transition between adjacent Landau levels. Cyclotron mass, $m^*_{cr\|}$ is related for the cyclotron energy, E_{cr}, by $E_{cr}=E_{LL}(\ell+1)-E_{LL}(\ell)=\hbar eB/m^*_{cr\|}$ and $E_{LL}(\ell)=(\ell+1/2)[\hbar eB/m^*_{LL\|}(\ell)]$ where $E_{LL}(\ell)$ is the energy of a specific Landau level, ℓ the Landau quantum number, e the electronic charge, and B is the magnetic field applied. By scanning B, we can change the energy, $E_{LL}(\ell)$, continuously. In principle, $m^*_{LL\|}(\ell)$ can be given by the differentiation of $E_{LL}(\ell)$ with B. In nonparabolic Landau levels, $m^*_{LL\|}(\ell)$ is

different at each ℓ. Then we can only measure the 'apparent' parallel mass, $m^*_{\mathrm{app}\|}$, given by

$$\frac{\mathrm{d}E_{\mathrm{cr}}}{\mathrm{d}B} = \hbar e \left(\frac{3}{2m^*_{\mathrm{LL}\|}(1)} - \frac{1}{2m^*_{\mathrm{LL}\|}(0)} \right) = \frac{\hbar e}{m^*_{\mathrm{app}\|}},$$

if the transition is the fundamental one.

4. Normal mass analysis

4.1. Calculation of eigenenergy and apparent normal mass

Theoretical eigenenergies were obtained from Yamada model [6]. They used the envelope function approximation based on Bastard's model [7], but all the eight bands were considered. Isolated QWs were assumed and wavefunctions decaying outside the well were used as shown in Eqs. (1)–(7) in Ref. [6]. Theoretical apparent normal effective mass was calculated from procedures given in Section 3.

4.2. Comparison of normal mass

Mass normal to the QW plane determines eigen-energies in a well. Eigenenergies are observed by spectroscopy of IR transmittance or photocurrent, due to the clear exciton-enhanced peaks and quantum number dependence [8]. Contribution of valence band was separable because valence-band parameters were already obtained experimentally [3]. In this experiment, we used photocurrent spectra taken at room temperature for V706 and V748 specimens. Each QW is assumed not coupled due to thick barrier and high band offset.

Experimental eigenenergies for specimens of V706 and V748 were almost proportional to the square of quantum number for quantum confinement as shown in Figs. 1 and 2. Because the energy plot seemed sublinear for both specimens, mass increment as a function of kinetic energy was expected.

Theoretical calculations using Yamada model fit very well with our experiments as shown in Figs. 1 and 2, where the shape of well potential was assumed to be square and the well thickness was 9, 10, and 11 nm for V706 and 20, 22, and 24 nm for V748. Using observed eigenenergies, experimental apparent normal mass of conduction electrons could be obtained by the procedure stated in Section 3.

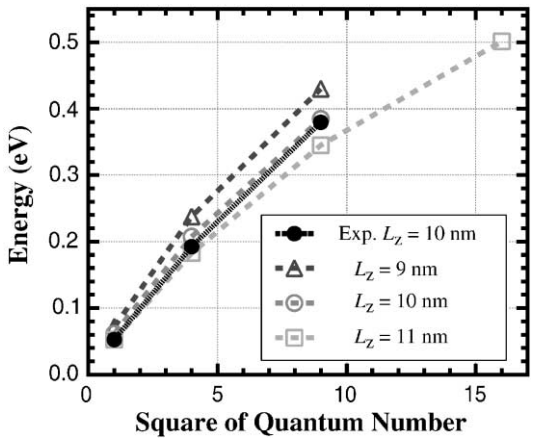


Fig. 1. Observed eigenenergies (solid circles) versus assigned quantum numbers squared, in a 10-nm-thick InGaAs QW (V706 specimen). Also shown are the calculated values (open marks) for 9- , 10- , and 11-nm-thick QWs.

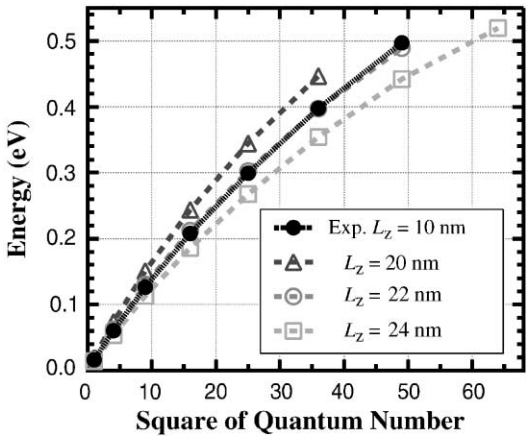


Fig. 2. Observed eigenenergies (solid circles) versus assigned quantum numbers squared, in a 20-nm-thick InGaAs QW (V748 specimen). Also shown are the calculated values (open marks) for 20- , 22- , and 24-nm-thick QWs.

The trend in mass increment as a function of energy agreed well with theory as shown in Figs. 3 and 4, where the central well thicknesses for both theory and experiment were assumed to be 10.5 and 22 nm, for V706 and V748 specimens, respectively. Measurement error in eigenenergies of ± 10 meV was also estimated in theory.

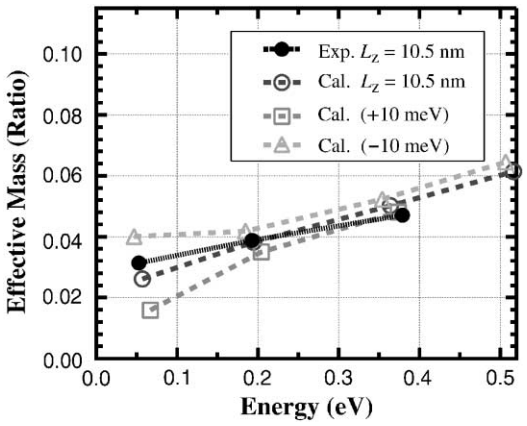


Fig. 3. Apparent normal mass (solid circles) of electrons observed in V706 specimen as a function of energy, where well thickness was assumed 10.5 nm. Calculated ones (open marks) were also shown.

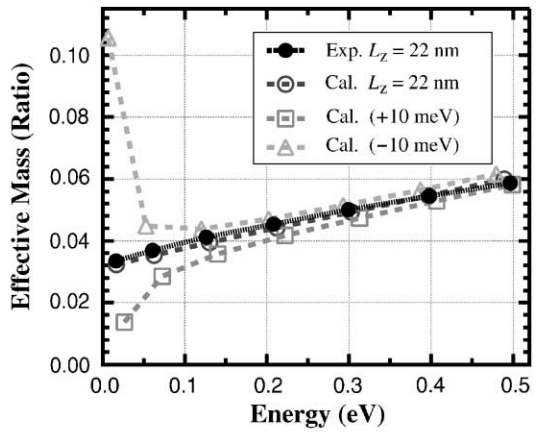


Fig. 4. Apparent normal mass (solid circles) of electrons observed in V748 specimen as a function of energy, where well thickness was assumed to be 22 nm. Calculated ones (open marks) were also shown.

5. Parallel mass analysis

5.1. Calculation of cyclotron energy and apparent parallel mass

Based on Zawadzki's model under Landau quantization, bulk Landau level was calculated from Eq. (14) of Ref. [9]. In QWs, the k_z^2 term was replaced using a relation, $\hbar^2 k^2/2m_\perp^*(E) = E_{\mathrm{QW}}(1)$, where $E_{\mathrm{QW}}(1)$ is the ground-state eigenenergy. The normal effective

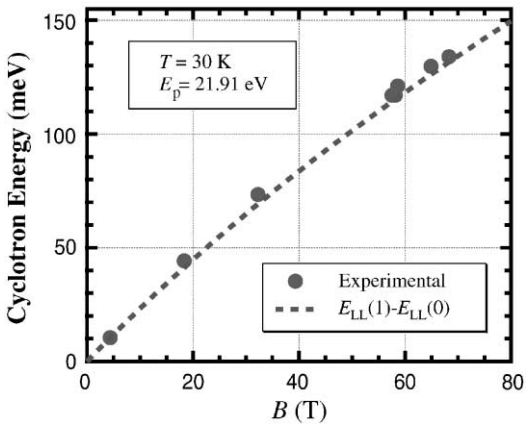


Fig. 5. Observed sublinear dependence of cyclotron energy in M10L specimen including 10-nm-thick InGaAs QW, as a function of magnetic field at 30 K. Also shown is the calculated cyclotron energy.

mass, $m_\perp^*(E)$, was already given in section 3. The difference in Landau level energies gave cyclotron energy, which gave apparent parallel mass, in theory.

5.2. Comparison of parallel mass

As seen in Fig. 5, measured cyclotron energy in M10L specimen was plotted as a function of B and compared with theory. The theoretical curve can be made to fit well with experiments by choosing a proper parameter, E_p, to be 21.91 eV for InGaAs, where $E_p = 2m_0M^2$ and M is the interband matrix element of the velocity operator in Kane's theory. This is the most probable one because it is the average of InAs and GaAs, 21.11 and 22.71 eV, respectively [10].

From this, all the Landau level energies for 10-nm-wide InGaAs QWs were revealed and the theoretical calculation of parallel mass became possible. In Fig. 6, the obtained apparent parallel mass was plotted by a solid line in the measured energy range. Bulk nonparabolic mass calculated from Kane's three-level theory was also compared and parallel mass was found closer to the bulk curve.

6. Conclusion

To conclude, experimental normal and parallel masses in InGaAs QWs were measured comprehen-

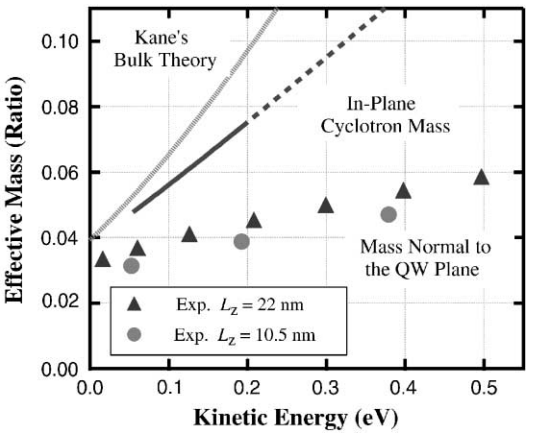


Fig. 6. Apparent parallel mass obtained from theory and experiment in M10L specimen including 10-nm-thick InGaAs QW, as a function of energy (thick solid line). Experimental apparent normal mass (solid marks) based on the photocurrent spectra are also shown for comparison.

sively. The apparent normal mass measured from photocurrent spectra agreed quantitatively with the model calculation proposed by Yamada et al., where a consistent E_p parameter was adopted. The experimental cyclotron energy agreed well with theory. The parallel mass was plotted by combining theory and experimental results.

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